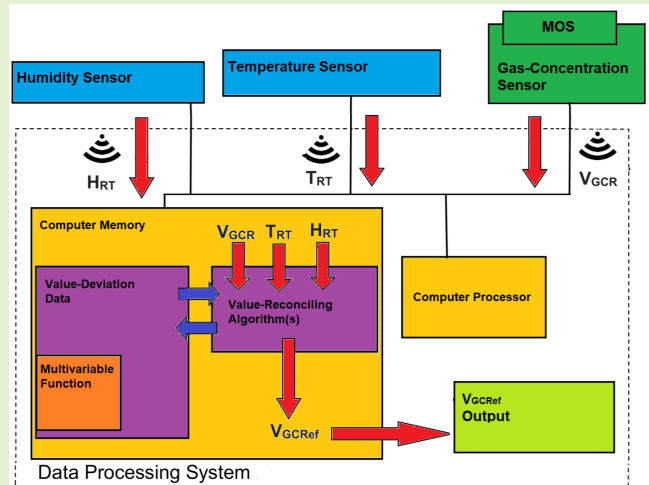


Algorithmic Correction of MOS Gas Sensor for Ambient Temperature and Relative Humidity Fluctuations

A. Aurora 

Abstract—In ambient conditions, uncontrolled fluctuations of ambient temperature and relative humidity significantly reduce the accuracy of metal oxide semiconductor-based gas sensors. The main innovation of the work is an algorithmic self-correction method that employs a response from an auxiliary temperature sensor and a multivariable polynomial regression model to minimize the effect of such environmental fluctuations. Tin oxide (SnO_2) MOS gas sensors were chosen for a sample study due to their fast response times and high resolution. Using the developed correction method, the study analyzes sensor accuracy while measuring a typical pollutant gas (carbon monoxide) over a temperature range from -10 to 40°C at relative humidity intervals of 35%, 65%, and 95%. On average, the corrected sensors measured a 40 ppm CO gas sample with 11 times more accuracy when compared to uncorrected sensors placed in the same variable ambient conditions. Corroborating ANOVA tests demonstrated that the prescribed correction algorithm reduced measurement variability with statistical significance. In the presence of CO_2 , CH_4 , ethanol, LPG, or smoke interference gas samples, sensors applying the correction method and temperature-based selectivity distinguished and measured CO concentrations with a maximum error of only ± 2.0 ppm. Thus, the proposed technical solution is attractive for highly variable environments and field applications where low-cost drift stabilization is necessary.

Index Terms—MOS gas sensor, ambient temperature, relative humidity, algorithmic correction, auxiliary temperature sensor.



I. INTRODUCTION

METAL oxide semiconductor (MOS) gas sensors employing tin dioxide (SnO_2) as the primary sensing material show promise for ambient air pollution monitoring due to many advantageous features such as high sensitivity, high selectivity, fast response and recovery, capability for low-cost mass production, and applicability in small-scale mobile applications [1]. Fundamentally, MOS gas sensors function by housing inexpensive sensing elements that react with target gases and produce a signal current in proportion to the gas concentration. In practice, the signal current is reflected by the magnitude of the instantaneous sensor resistance – a value that can be mathematically converted to a gas concentration in ppm [2].

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Tin dioxide (SnO_2) is an n-type wide bandgap semiconductor sensing element that has received attention in gas sensing due to its tunable physicochemical properties. The morphable nanostructures, high electron mobility, and high positive conduction band potential of tin dioxide mean the compound is versatile for detecting low gas concentrations with high sensitivity [3], [4]. The sensor technology has been shown to measure various volatile organic compounds (VOCs), NO_2 , CO , CH_2 , and H_2S gas with reasonable precision and accuracy [5]–[7]. As a result, SnO_2 MOS gas sensor technology has been widely applied in automatic ventilation control systems, household applications, and recently in toxic gas sensor devices [8]. However, many drawbacks exist restricting the use of MOS technology more widely in ambient environments for pollutant gas sensing. These include frequently observed drift, poor quality due to reproducibility challenges, narrow dynamic range, and non-linear response [9], [10]. The main consequences of these drawbacks are poor accuracy and precision, especially in field applications where measurement conditions are highly variable.

Uncontrolled fluctuations of ambient temperature and relative humidity are the two prime environmental factors that significantly reduce the accuracy of MOS gas sensors [11]–[14]. Previous research efforts have focused on developing techniques for direct temperature compensation to eliminate bias from gas sensor responses. One approach is to integrate temperature feedback control circuits to regulate the sensor microheater's current and maintain a constant working temperature, regardless of ambient temperature change [15]. Another proposed method dopes the semiconductor material with micromachined substrates to amplify the thermal sensitivity of the sensing material and utilize thermal selectivity to determine the gas concentration [16].

While these techniques can be applied in indoor environments with low variations in temperature and relative humidity [17], in outdoor environments with significant hourly variations, these methods fail to compensate effectively [18], [19]. For this reason, physical and chemical solutions tend to have limitations depending on the degree of environmental change, and they also significantly increase gas sensor costs. Therefore, this study proposes an analytical method to improve MOS sensor accuracy by quantifying the impact of temperature and relative humidity change and correcting the sensor response in real-time.

In summary, this research provides a characterization of MOS gas sensor behavior as ambient temperature and relative humidity change simultaneously. Based on these results, a novel method combining an auxiliary temperature/relative humidity sensor's response with a self-correction algorithm is presented to mitigate sensor resistance drift. And utilizing this self-correction algorithm, sensor performance was assessed in carbon monoxide gas samples over the temperature range from -10°C to 40°C at relative humidity intervals of 35%, 65%, and 95% for thirty randomly selected Figaro TGS2600 sensors.

The objective of the paper is to provide a low-cost method of enhancing tin oxide-based gas sensor accuracy to enable their application for ambient air pollution monitoring operations. Three aspects of sensor performance will be studied in the subsequent sections: (1) the correlation of sensor resistance change to various temperature and relative humidity conditions, (2) the improvement of sensor response accuracy by active algorithmic correction, and (3) the assessment of the correction algorithm's efficacy in the presence of interference gasses.

II. MATERIALS AND METHODS

A. Resistance Versus Ambient Temperature/Relative Humidity Curve Calculation

The sensor mechanism of a typical MOS gas sensor is shown in Fig. 1.

Where V_C is the circuit voltage (in volts), R_L is a load resistor (in Ohms), V_{RL} is the voltage across the load resistor (in volts), V_H is the heater voltage applied to the integrated heater in order to maintain the sensing element at a specific temperature which is optimal for sensing (in volts), R_H is the heater circuit resistor (in Ohms), and R_s is sensor resistance (in Ohms).

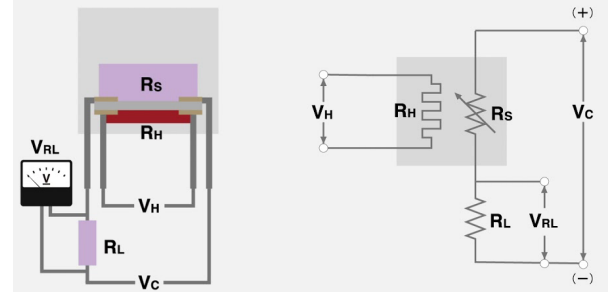


Fig. 1. Metal oxide-semiconductor gas sensor mechanism and circuit schematic [20].

MOS gas sensor resistance (R_s) is calculated from a measured value of V_{OUT} (V_{RL}) by using the following formula:

$$R_s = \left(\frac{V_C}{V_{RL}} - 1 \right) R_L \quad (1)$$

One method of obtaining gas sample concentration algorithmically from the R_s value is by conducting power regression on the sensitivity characteristic parameters of the sensor for different gasses provided by the manufacturer.

Relating the sensor resistance ratio (R_s/R_0) to the sample gas concentration yields the following formula:

$$C_{gs} = a \left(\frac{R_s}{R_0} \right)^b \quad (2)$$

where R_0 is sensor resistance in a sample of fresh" air at 20°C and 65% relative humidity (in Ohms), C_{gs} is the concentration of the gas sample (ppm), and [a, b] are coefficients specific to the gas chemistry.

Employing this formula, a temperature/humidity dependency curve can be constructed, and the variability of the sensor resistance ratio over a chosen temperature and humidity range can be related. Note that variability and weak reproducibility in R_s response between individual sensors exists inherently due to the manufacturing process [21]. Therefore, a more accurate representation of sensor response is R_s/R_0 which is independent of variation among sensors for standard resistance readings.

Modeling the resistance ratio correction curve enables the generation of a sensor resistance correction factor, K , when combined with the response from an auxiliary temperature sensor. To yield corrected concentration values, the value of K can be integrated into the gas concentration calculation equation as shown (3).

$$C_{gs} = a \left(\frac{R_s}{R_0 K} \right)^b \quad (3)$$

Accurate gas concentration measurements can be obtained by applying this method through a computer-programmed algorithm.

B. Experimental Apparatus

The main components are shown in Fig. 2, with a schematic representation of the experimental gas measurement instrument in Fig. 3. The apparatus consists of a climate

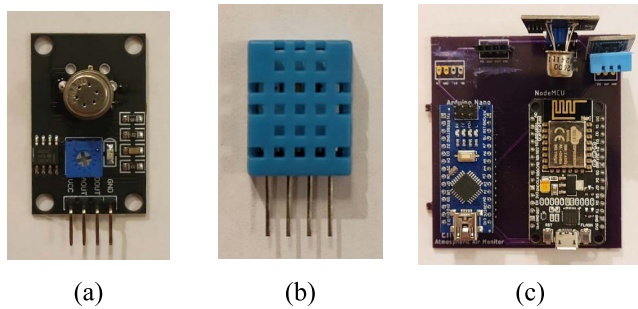


Fig. 2. Images of the key experimental components: (a) Figaro TGS2600 sensor, (b) DHT11 sensor, (c) PCB mounting sensors and wiring to Arduino Nano and NodeMCU.

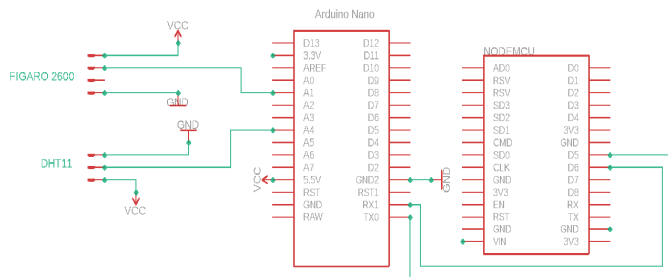


Fig. 3. Schematic representation of the gas sensor complex used for experimentation.

chamber [22], a Figaro TGS2600 gas sensor [20], a DHT11 temperature and humidity sensor [23], an Arduino Nano micro-controller [24], and a NodeMCU Wi-Fi module [25].

C. Climate Chamber

The experiment occurred in a Wiess Technik climate-controlled chamber (WK180/40) programmed for a stepped temperature range from -10°C to 40°C stabilizing in 5°C intervals at relative humidities of 35%, 65%, and 95%. At each interval, there are approximately 20 minutes of stable measurement time. With an accuracy of $\pm 0.1^{\circ}\text{C}$ and $\pm 5\%\text{RH}$, the climate chamber was calibrated at two temperature reference points (25°C and 60°C) and two relative humidity reference points (20% and 95%). The humidity was kept constant as the temperature changed by using an evaporator and cooling coil connected to the climate chamber. When humidity was lower than the desired level, water from an external tank was heated up in the evaporator and fed as steam into the inner chamber. When humidity was higher than the desired level, a cooling coil in the chamber attracted moisture in the air, causing it to condense into water, which was then drained away. A digital sensor (SHT 75, Sensirion, Zürich, Switzerland) measured the temperature and humidity inside the climate chamber. The manufacturer reports the accuracy of the sensor for humidity and temperature is $\pm 1.8\%$ and $\pm 0.3^{\circ}\text{C}$, respectively. Climate chamber sensor responses were monitored and stored using a real-time data logging program developed at DPI&F, Queensland using Labview 7.1™ (National Instruments, Austin, TX, USA). Five-second measurement intervals of temperature and relative humidity by the SHT 75 sensor enabled high-accuracy, controlled humidification.

The climate chamber housed a polyethylene gas chamber with the measurement instrument.

D. Gas Sensors

Thirty randomly selected pre-calibrated Figaro TGS2600 MOS-based sensors were studied (Fig. 3a). The TGS2600 sensor provides high-resolution gas concentration data for CH_4 , CO , iso-butane, ethanol, and hydrogen gas. A power regression model based on the manufacturer-provided calibration curves was employed to calculate experimental CO concentrations. The sensors communicate with the micro-controller via analog pins under standard circuit conditions at a serial communication rate of 9600 baud.

E. Auxiliary Temperature and Humidity Sensor

Thirty randomly selected DHT11 sensors (Fig. 3b) provided the auxiliary temperature and humidity data necessary for the correction algorithm. The sensors have an accuracy of $\pm 2^{\circ}\text{C}$ and $\pm 5\%\text{RH}$ with 16-bit resolution. The sensors communicate with the micro-controller via analog pins under standard circuit conditions at a serial communication rate of 9600 baud.

F. Microcontroller

An Arduino Nano processes the incoming data from the gas sensor and integrates a correction algorithm to calculate CO concentration (in ppm). A NodeMCU Wi-Fi module connected to the Arduino Nano stores and displays uncorrected, corrected, and sensor resistance data in a MATLAB ThingSpeak channel.

G. ThingSpeak Channel

The ThingSpeak channel facilitated data storage and visualization. The primary data collected were sensor resistance (R_s), sensor resistance ratio (R_s/R_0), gas concentration (C_{gs}), ambient temperature (T), ambient relative humidity (R_H), and the correction factor (K). All dependent variables were graphed with time as the independent variable.

H. Sensing Chamber

A leak-proof 3.875-liter polyethylene box encased each of 20 gas sensor complexes, forming the sensor array. To route wires connecting the sensor array to a power outlet, a makeshift access port was cut into the sensing chamber. The sensing chamber also has a valve connecting directly to a gas conditioning unit (GCU) to allow entry of gas samples. All external connections were leak-proofed with epoxy plastic.

An experimental method proposed by Sohn *et al.* was applied to test gas sensor accuracy and qualify the devised algorithm [13].

I. Experimental Procedure

To determine the correlation between temperature/humidity change and the sensor resistance ratio, “fresh” air samples were introduced to the sensing chamber at a 200 mL/min constant flow rate controlled by a mass flow controller (Aalborg® Instruments & Controls Inc., New York, USA).

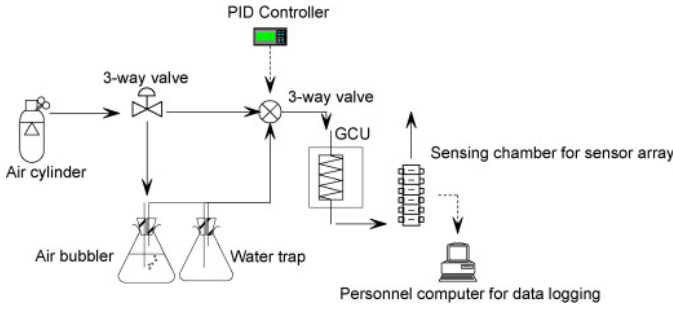


Fig. 4. Schematic of the experimental setup to release gas sample into the chamber [13].

The sensor array was exposed to the sample for 1800 seconds at each trial utilizing the GCU. The GCU was designed for (1) pre-conditioning gas samples at specific temperatures and humidities, (2) providing gas samples to the sensing chamber of the gas sensor array, and (3) controlling gas flow rate and pressure.

The MOS sensor array was warmed up for 3600 seconds to stabilize the sensors' responses at equilibrium before collecting the analyzed sampling data. Dry air (5% relative humidity) from a gas cylinder was then pumped into the chamber for 1800 seconds to purge the whole system and remove any residuals from previous tests. Finally, after being properly preconditioned by the GCU, the gas sample was introduced to the gas chamber inside the climate chamber.

The first 600 seconds of data were discarded to analyze only fully stabilized sensor outputs. Averaged measurements from the twenty-second stabilization intervals were collected from the remaining 1200 seconds of data to minimize the effects caused by white and colored noises generated from electronics and sensors.

To test the correction algorithm-enabled sensors' accuracy improvement in a practical application, they were compared to uncorrected sensors in the presence of 40 ppm of CO. The above procedure was replicated but with the prescribed CO gas mix, and each sensor complex's response (including the gas sensor and temperature/humidity sensor) was measured to yield the uncorrected dataset and then the correction algorithm was applied to yield the corrected dataset.

To test the correction algorithm-enabled sensors' accuracy improvement in the presence of a target gas and other interference gasses, the sensors' responses were recorded in a gas mix of 40 ppm CO and 5 ppm of CO₂, CH₄, ethanol, LPG, or smoke. The above procedure was replicated, but a circuit voltage feedback loop was employed to maintain a working temperature of 250°C. Sensor responses were compared to the target gas concentration to qualify the correction algorithm's ability for reliable gas selectivity.

For each MOS gas sensor and auxiliary sensor, 33 data points were collected during each trial. And over a testing period of 9 months, 19800 gas sensor responses were collected from 20 trials in "fresh" air, 19800 gas sensor responses were collected from 20 trials in the 40 ppm CO gas mix, and 3960 gas sensor responses were collected from 4 trials for each of the 5 gas mixes with 40 ppm CO and 5 ppm of an interference gas. The schematic diagram of the adapted gas chamber setup is presented in Fig. 4.

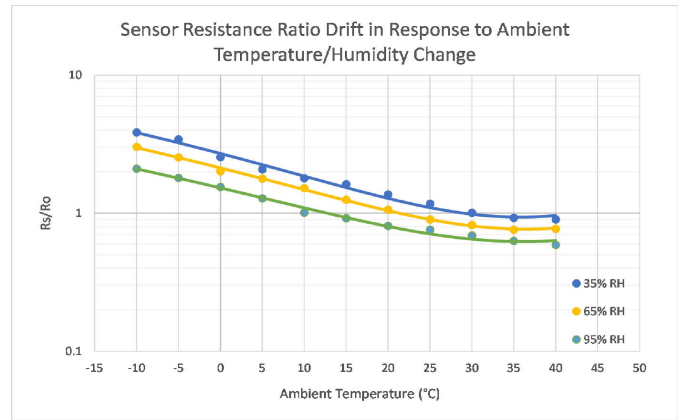


Fig. 5. Mean sensor resistance ratio drift over a temperature range from -10°C to 40°C at relative humidities of 35%, 65%, and 95% with second-degree polynomial curve trendlines (solid line).

III. RESULTS

A. Modeling the Impact of Ambient Temperature and Relative Humidity

The accuracy of the Figaro TGS2600 sensors' behavior was investigated over the previously described temperature and relative humidity range to determine a trend in sensor resistance ratio change.

The experimental R_s/R_0 response curves for all gas sensors in "fresh" air at variable temperature and relative humidity are presented in Fig. 5, showing the negative relationship between sensor resistance and temperature at 35%, 65%, and 95% relative humidity. To produce the displayed graph and dataset, R_s/R_0 values for all sensors at each temperature-humidity pairing were averaged. Fig. 5 also shows a fit to second-order polynomial curves for sensor resistance ratio change versus climate chamber temperature measurements at variable relative humidity. Only data recorded within the plateau regions of intervals were analyzed for gas sensor resistance. Although chamber leakage was negligible, the polyethylene warranted a carbon monoxide concentration decay over long periods of time, restricting the testing session length at each temperature and relative humidity interval.

The symmetric nature of the curves with vertical transformations suggests a truncation of the multivariate Taylor series to a second-order polynomial for modeling the relationship between sensor resistance ratio and temperature.

The second-order multivariable polynomial curve form is presented in (4) along with the extracted experimental mean coefficients in Table I.

$$K(T, R_H) = aT^2 + bR_H^2 + cTR_H + dT + eR_H + f, \\ \times \{(T, R_H) | -10 \leq T \leq 40, 35 < R_H \leq 95\} \quad (4)$$

where, a , b , c , d , e , and f represent the coefficients that model the mean dependence of R_s/R_0 upon ambient temperature and relative humidity. The obtained values in Table I reflect a high coefficient of determination ($r^2 = 0.9873$) [26].

TABLE I
BEST-FITTING MULTIVARIABLE POLYNOMIAL COEFFICIENTS

Coefficients	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>
Mean curve	0.00103	$-3.101 \cdot 10^{-6}$	$4.698 \cdot 10^{-4}$	-0.105	-0.0196	3.404

Coefficients of the best-fitting second-order multivariable polynomial to model mean sensor resistance ratio drift under variable temperature/humidity conditions.

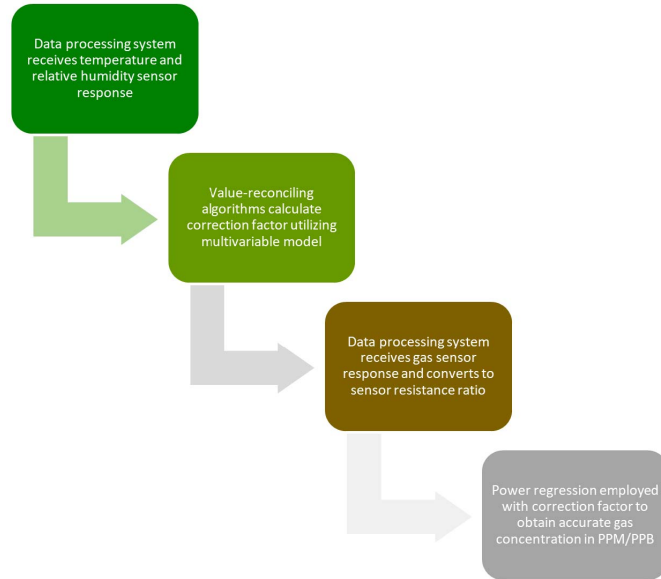


Fig. 6. Code logic for the correction algorithm using multivariable polynomial modeling.

B. Application of the Polynomial Model to Algorithm

The multivariable polynomial model was linked to the gas concentration measurement power regression model and integrated into an algorithm. Programmed in C/C++, the algorithm was uploaded to the Arduino Nano microcontroller for data processing. The basic algorithm code logic is presented in Fig. 6.

C. Carbon Monoxide Testing

To demonstrate the application of the self-correction method for measuring a typical pollutant gas, sensor resistance and gas concentration data was collected for sensors placed in a 40 ppm CO gas sample with the previously selected temperature and humidity range. Corrected responses were compared to uncorrected responses in the same environments to quantify and prove the self-correction method's accuracy. Mean CO gas concentration measurements for corrected and uncorrected sensors are shown in Fig. 7.

For uncorrected sensors, deviation in measured CO concentration increases as temperature exits the $20 \pm 5^\circ\text{C}$ range. Detected CO concentration increases relatively linearly for uncorrected sensors as temperature increases. The correlation between temperature and CO concentration appears consistent across relative humidity conditions. However, there is a clear vertical transformation evident such that as relative humidity increases, CO concentration also increases.

Corrected sensor response deviates from the target concentrations minimally and predictably. The drift between samples

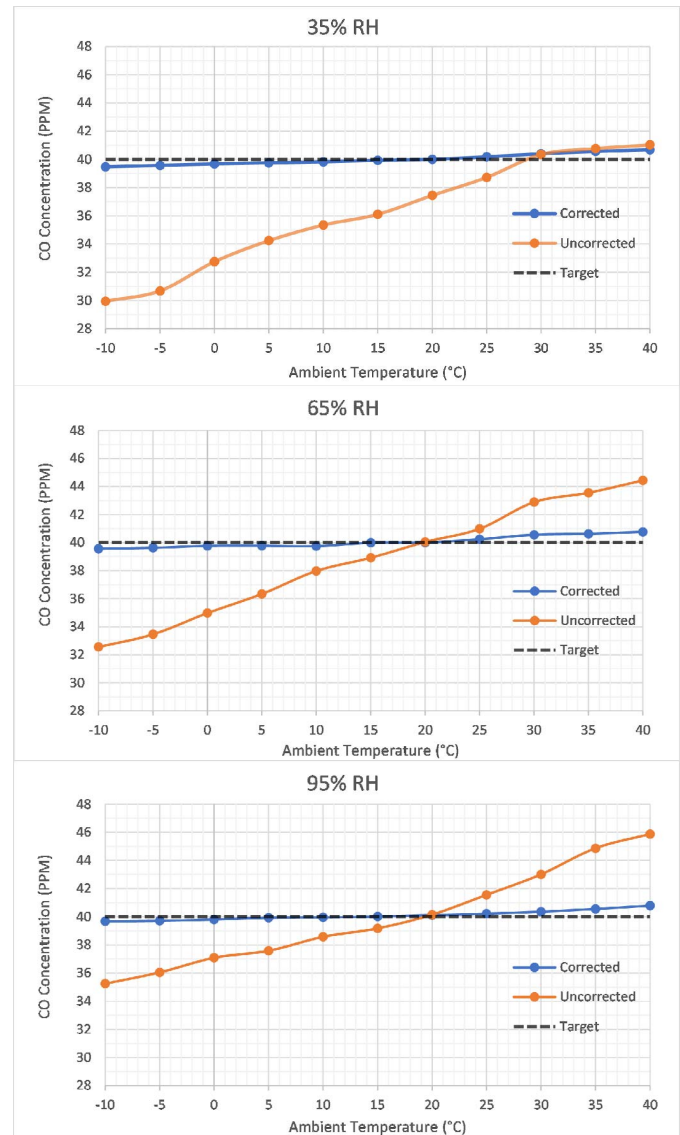


Fig. 7. Gas sensor response with and without the correction algorithm in 40 ppm CO under variable temperature/humidity conditions.

due to different relative humidity conditions seems negligible. However, the same deviation pattern is observed for corrected sensors as for uncorrected sensors when the temperature changes. Measured CO concentration for corrected sensors only noticeably deviates from 40 ppm when temperature exits the $20 \pm 5^\circ\text{C}$ range.

The absolute residual concentration calculation errors for corrected and uncorrected sensors are presented in Fig. 8.

The difference in the deviation from the target measurement between corrected and uncorrected samples demonstrates the magnitude of sensor accuracy improvement after correction. The data points represent the mean value of the absolute residual error for all sensors at 35%, 65%, and 95% relative humidity. The accuracy of all correction algorithm-enabled sensors is within 0.8 ppm. However, the accuracy of uncorrected sensors ranges up to 12 ppm. The specific absolute value residual means and variances for uncorrected and corrected sensors at each controlled relative humidity are given in Table II.

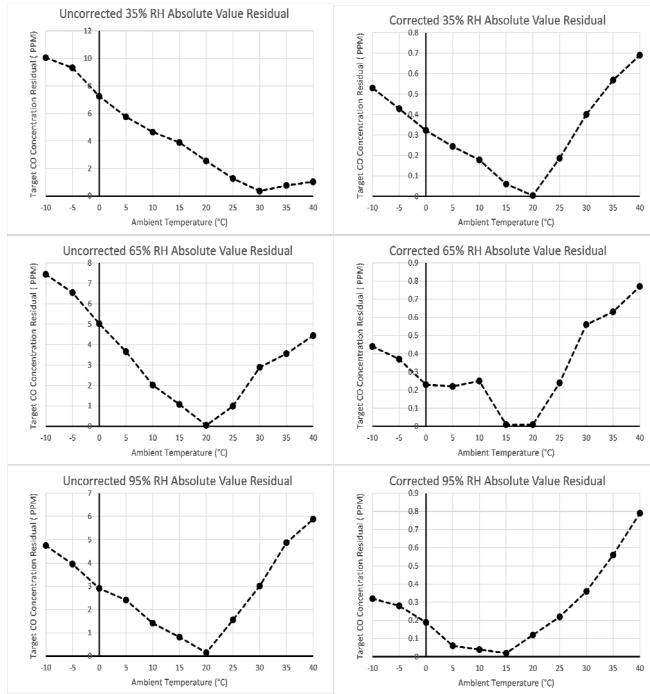


Fig. 8. Residual error with and without the correction algorithm in 40 ppm CO under variable temperature/humidity conditions. Note the scale differences in target CO concentration for corrected and uncorrected samples.

TABLE II
ABSOLUTE RESIDUAL MEAN AND VARIANCE

Sensor Type	Parameter	35%	65%	95%
Corrected	Mean	0.328	0.339	0.269
	Variance	0.0471	0.059	0.056
Uncorrected	Mean	4.264	3.426	2.885
	Variance	12.038	5.455	3.341

Means and variances of absolute residuals for corrected and uncorrected gas samples at 35%, 65%, and 95% relative humidity.

On average, the mean absolute residual is 11 times less for corrected samples than uncorrected samples, representing a substantial performance improvement. Variance within the various humidity samples also decreased significantly in corrected samples compared to uncorrected samples.

D. ANOVA Testing

To test the statistical significance of the correction method, an analysis of variance test was conducted for the absolute value residuals of the corrected and uncorrected samples. Test results for 35%, 65%, and 95% relative humidity are presented in Table III.

ANOVA test results show that for all observed relative humidity conditions, F-statistics were higher than 1 at p-values lower than $\alpha = 0.05$. Thus, the difference in variability between the corrected sample and the uncorrected sample for absolute value residuals is significant.

E. Correction-Enabled Selectivity Testing

The effect of interference gasses on the correction-enabled sensors' accuracy was investigated by applying the Carbon

TABLE III
ANOVA TESTING

Relative Humidity	35%	65%	95%
F-statistic	22.153	19.014	14.099
p-value	0.000135	0.000303	0.00125

F-statistics and p-values for ANOVA tests of corrected and uncorrected gas sensor samples at 35%, 65%, and 95% relative humidity.

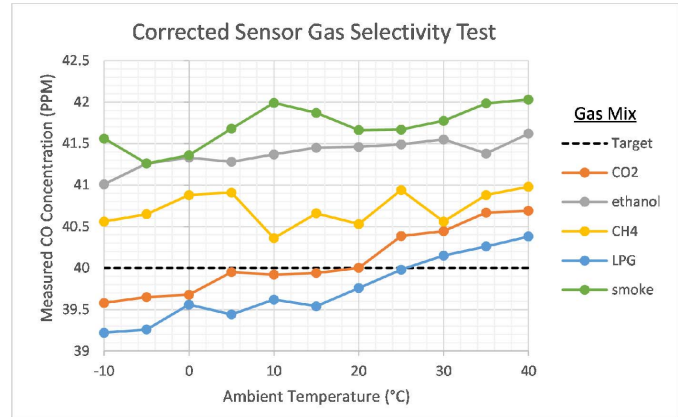


Fig. 9. Mean corrected gas sensor response in various gas mixes with 40 ppm CO and 5 ppm of an interference gas.

Monoxide Testing procedure to five separate gas mixes containing 40 ppm of CO and 5 ppm of either CO₂, CH₄, ethanol, LPG, or smoke. Each Figaro TGS2600 unit was operated at 250°C for sensor heater temperature-based gas selectivity of CO [5]. Mean corrected sensor responses for all gas mixes are shown in Fig. 9.

Despite the presence of interference gasses, the mean corrected gas sensor response maintains an accuracy of approximately ± 2.0 ppm. The mean bias of LPG, CO₂, and CH₄ is minimized to values below ± 1.0 ppm, but ethanol and smoke have a considerable effect on accuracy with a mean bias upwards of +1.0 ppm.

IV. DISCUSSION

In atmospheric gas concentration measurement operations, accuracy requirements for a precise measurement will depend on the application. However, the proposed self-correction method proves both versatile and effective given its low cost, considerable accuracy improvement, and adaptability to other MOS gas sensor technologies.

Additionally, this study develops an analytical methodology for coping with sensor drift without altering sensor properties. Such an approach is facilitated by the discovery that the sensor resistance drift due to ambient temperature and relative humidity change is predictable, albeit non-linear. And by measuring these environmental conditions with an auxiliary sensor, the MOS gas sensor resistance ratio can be adjusted in real-time to improve measurement accuracy.

A. Relationship Modeling

The gas concentration measurement accuracy of a MOS gas sensor, such as the one studied in this paper (Figaro TGS2600),

heavily depends on ambient temperature and relative humidity range. This is aptly demonstrated by the observed relationship between the sensor resistance ratio and the controlled temperature/relative humidity change (Fig. 5). All the dependency curves show similar trends in resistance ratio drift, echoing the replicability of Figaro TGS2600 response across the variable ambient factors. Finally, the dependency curves are also predictable and can be confidently modeled with a second-order multivariable polynomial curve.

B. Accuracy Improvement

In the CO gas experiment, corrected sensors consistently had lower means and variances of absolute value residuals as compared to uncorrected sensors (Table II). Corresponding ANOVA testing also proved that the accuracy improvement of the correction method is statistically significant. In practice, this means that modeling and adjusting sensor resistance ratio drift effectively accounts for changes in ambient conditions while measuring concentrations of an isolated target gas. The advantage of this “accounting” method lies in its versatility. Provided with the dependency curve and ambient conditions at the time of sensor operation, any MOS gas sensor’s response can be revised for the effects of environmentally induced drift. Thus, the method is universally applicable, even without the construction of the gas sensor complex used in the experimental apparatus; the gas sensor response dataset and auxiliary sensor response dataset will suffice for accurate correction.

To consider a more practical case in the field, the CO gas experiment was repeated with interference gasses. And as demonstrated, sensors employing temperature-based selectivity and the correction algorithm also measured CO concentrations with high accuracy in 40 ppm CO and 5 ppm of an interference gas.

In selectivity testing (Fig. 9), bias due to CH₄, ethanol, and smoke (containing CO and trace VOCs) was likely higher because they are target gasses for SnO₂ at operating temperatures closer to 250°C. Thus, when ambient temperature increased, the measured CO concentration increased as the gas sensors’ sensing elements approached higher operating temperatures, and adsorption occurred with compounds other than CO. The ideal operating temperature for LPG and CO₂ is lower than that of CO, thus the bias of these interference gasses is relatively low [27]–[30].

In any case, the selectivity tests demonstrate that controlling heater temperature through circuit manipulations affords a low-cost method of selectivity that warrants significant measurement accuracy when combined with the prescribed correction method. Such results confirm MOS gas sensor’s potential for application in conventional field environments polluted by several gas types.

C. Limitations

Uncorrected and corrected residuals tend to be higher at higher temperatures, especially as relative humidity rises (Fig. 8). This is likely due to the heating of the sensor TO-5 container package, which is not accounted for by the auxiliary temperature sensor and therefore may produce an

underestimated correction factor as ambient temperature and relative humidity rise.

Extensions of the correction method to other gas types may present difficulty if the physical characteristics of the target gas change as ambient temperature and relative humidity rise. Thus, the margin of error for measuring other gasses may be lower or higher depending on the target gas chemistry.

Due to practical issues restricting the utilization of smaller gas samples for carbon monoxide gas testing, a 40 ppm sample was chosen. However, this concentration may be unrepresentative of the lower concentrations found in many ambient atmospheric environments. As such, conducting the prescribed experiment at a broader range of CO concentrations with more granularity of temperature and relative humidity concentrations may provide more insight into the algorithm’s applicability to other environmental conditions. Additionally, trials with MOS gas sensors from different manufacturers may necessitate considerations of sensing material characteristics when applying the presented method. Finally, applying the methodology to other pollutant gasses may help solidify a more practical, multipurpose use of MOS gas sensors in ambient environments.

V. CONCLUSION

This study presents an ambient temperature and relative humidity correction technique for MOS-based gas sensors founded upon second-order multivariate polynomial modeling and an auxiliary temperature sensor’s response. The established method improves the accuracy of Figaro TGS2600 modules statistically significantly over a wide range of temperatures (−10°C to 40°C) and relative humidities (35%, 65%, and 95%). Carbon monoxide testing in a 40 ppm gas sample demonstrated that sensor accuracy improves by up to 11 times with the correction algorithm and reduces the error to approximately ± 0.8 ppm CO. In the presence of CO₂, CH₄, ethanol, LPG, and smoke gas interference samples, combining the self-correction method with temperature-based gas selectivity enables relatively high accuracy target gas measurements, limiting error to ± 2.0 ppm CO. Thus, MOS-based gas sensors employing this method for adapting to ambient environments are able to retain gas concentration measurement integrity. When applied, the developed drift mitigation approach empowers MOS gas sensor usage in outdoor applications for low-cost yet high-accuracy pollutant gas sensing.

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